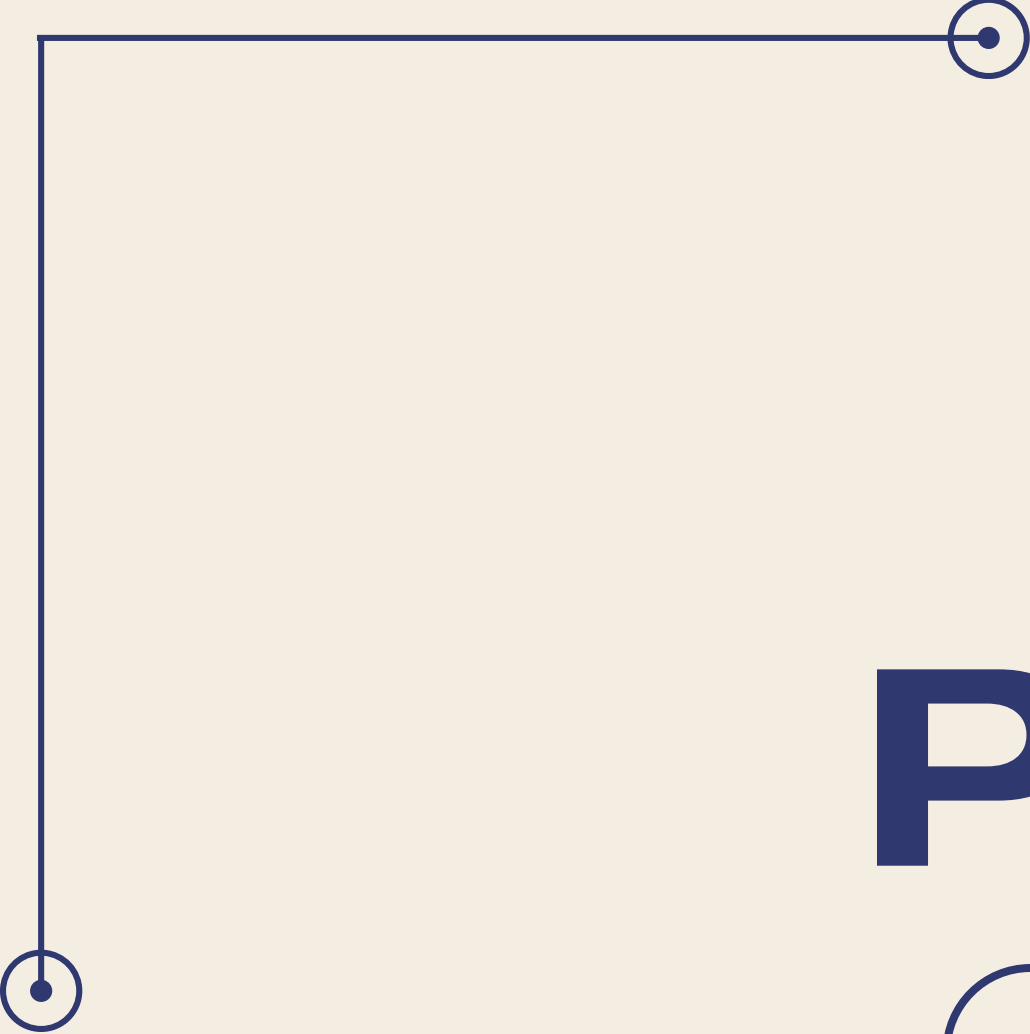
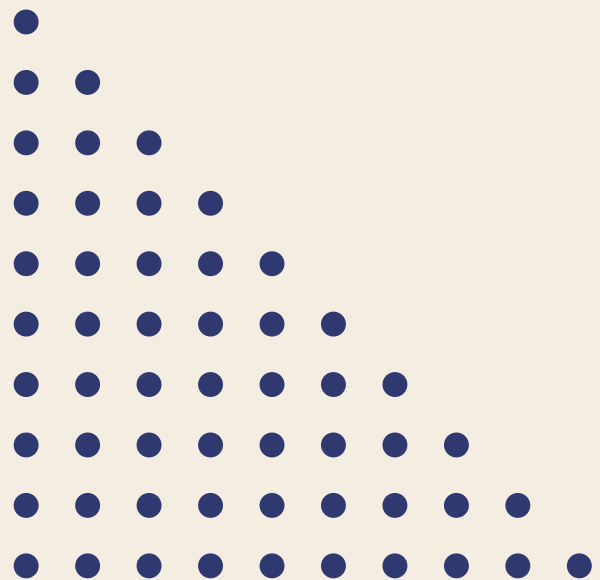




PROBLEM-7

PCAP Forensics and Attack Isolation



OUR TEAM

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Document Configuration, IP
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OUR CONTENTS

01. Problem Scenario

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PROBLEM SCENARIO

SITUATION: A homeowner's smart devices started acting strange. Lights turned on/off by themselves. Internet was slow.

OUR TASK: Analyze the PCAP file to find the truth.

THE QUESTIONS:

- Which device was hacked?
- Who is the attacker?
- How did they break in?
- How do we stop it?

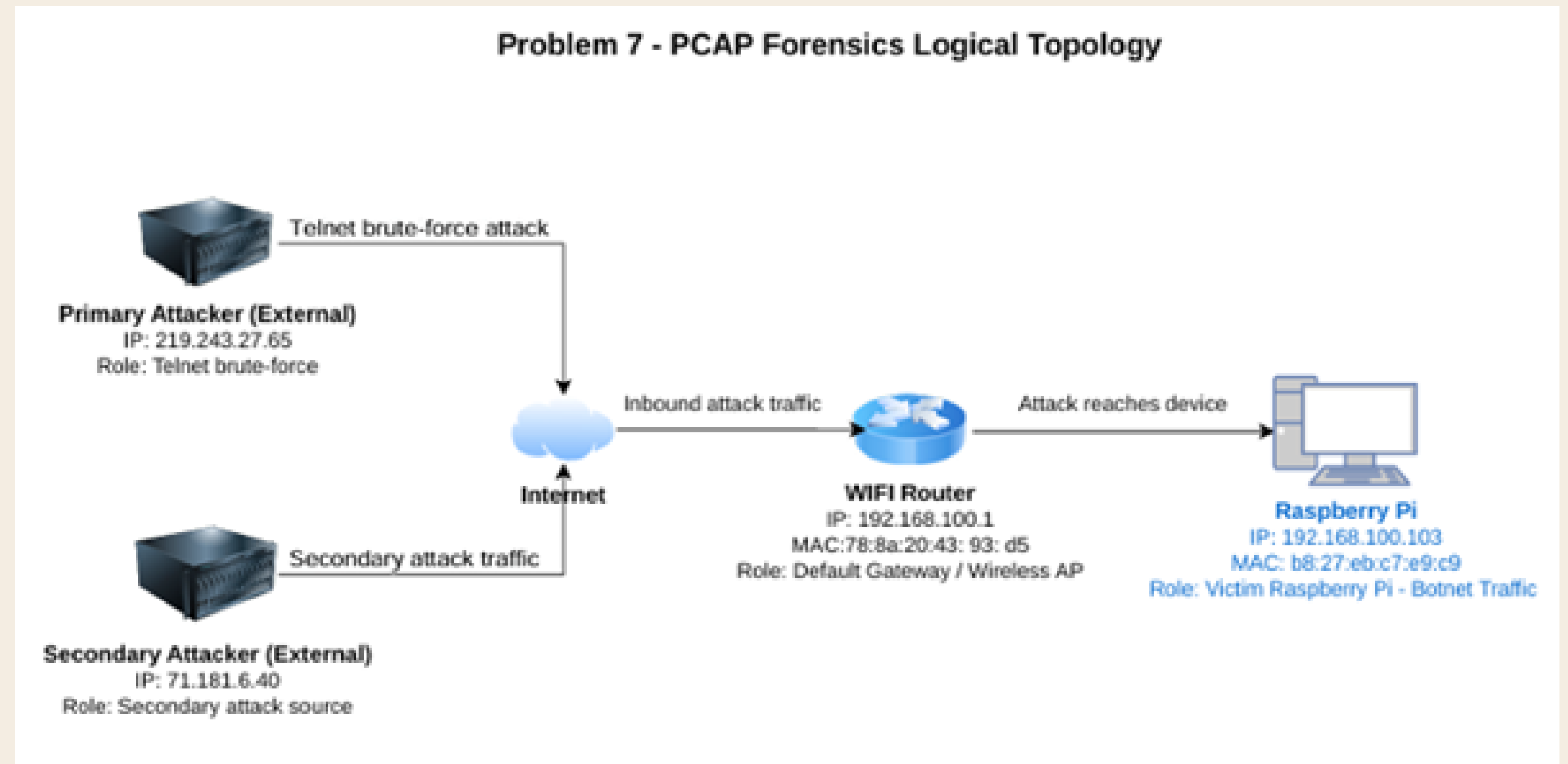
METHODOLOGY

- 1 PCAP file loaded in Wireshark
- 2 MAC address filtering to identify compromised device
- 3 IP conversation analysis to find C2 server
- 4 Protocol filter for brute force detection
- 5 GNS3 topology built for isolation & mitigation

NETWORK DIAGRAM

Our network has these devices:

- Internet (Cloud) - The outside world
- Primary Attacker (219.243.27.65) - Main C2 server
- Secondary Attacker (71.181.6.40) - Backup C2 server
- WiFi Router (192.168.100.1) - Home gateway
- Raspberry Pi (192.168.100.103) - Hacked device (victim)



DEVICE IDENTIFICATION

DID

We identified the compromised device by analyzing the PCAP file in Wireshark



FINDINGS

- Device: Raspberry Pi
- IP: 192.168.100.103
- MAC: 58:27:eb:c7:e9:c9
- Vendor: Raspberry Pi Foundation
- Status: Compromised
- Attack Role: Botnet Scanner

EVIDENCE 1

2018-05-09-192.168.100.103.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.100.103	154.8.94.65	UDP	60	43763 → 14336 Len=12
2	0.000738	192.168.100.103	65.127.233.163	TCP	74	51524 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178361 TSecr=0 WS=64
3	0.009982	192.168.100.103	63.150.16.171	TCP	74	56305 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178362 TSecr=0 WS=64
4	0.029972	192.168.100.103	111.40.23.49	TCP	74	41101 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178364 TSecr=0 WS=64
5	0.043964	Ubiquiti_43:93:d5	RaspberryPiF_c7:e9:...	ARP	60	Who has 192.168.100.103? Tell 192.168.100.1
6	0.044462	RaspberryPiF_c7:e9:...	Ubiquiti_43:93:d5	ARP	60	192.168.100.103 is at b8:27:eb:c7:e9:c9
7	0.999928	192.168.100.103	65.127.233.163	TCP	74	[TCP Retransmission] 51524 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178361 TSecr=0 WS=64
8	1.000420	192.168.100.103	196.170.198.141	UDP	60	43763 → 11764 Len=12
9	1.001167	192.168.100.103	131.174.215.147	TCP	74	60905 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178461 TSecr=0 WS=64
10	1.009912	192.168.100.103	91.42.47.63	TCP	74	44301 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178462 TSecr=0 WS=64
11	1.029902	192.168.100.103	120.210.108.200	TCP	74	50244 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178464 TSecr=0 WS=64
12	1.999868	192.168.100.103	131.174.215.147	TCP	74	[TCP Retransmission] 60905 → 23 [SYN] Seq=0 Win=29200 Len=0 MSS=1460 SACK_PERM TSval=182178461 TSecr=0 WS=64
13	2.000351	192.168.100.103	118.112.204.139	UDP	60	43763 → 2441 Len=12

> Frame 1: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits)
> Ethernet II, Src: RaspberryPiF_c7:e9:c9 (b8:27:eb:c7:e9:c9), Dst: Ubiquiti_43:93:d5 (7)
> Internet Protocol Version 4, Src: 192.168.100.103, Dst: 154.8.94.65
> User Datagram Protocol, Src Port: 43763, Dst Port: 14336
> Data (12 bytes)

Ethernet

Destination

Source

Type

Padding

2018-05-09-192.168.100.103.pcap

Packets: 1686291

Profile: ariba

C2-SERVER COMMUNICATION

(ATTACKER CONTROL)

DID

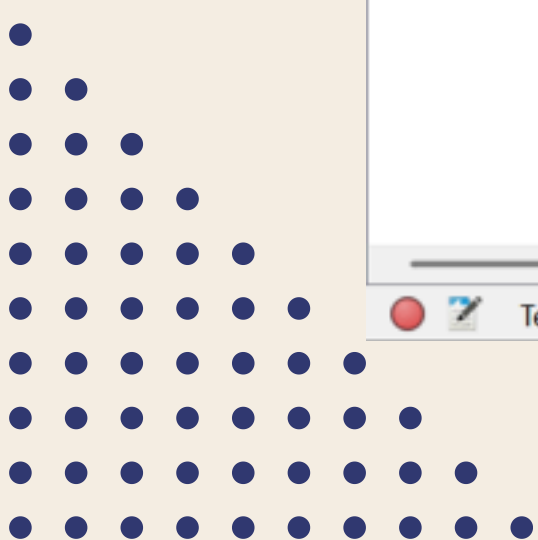
The attacker's server (219.243.27.65) sent many Telnet packets to the hacked device (192.168.100.103)

FINDINGS

- Attacker IP: 219.243.27.65
- Used Telnet (Port 23)
- Packet size: 60 to 236+ bytes
- Non-stop data flow

WHY IS IT CRITICAL

The attacker is controlling your device from far away. He can send any command, steal your information, or use your device to attack others. Your device is not yours anymore.



BRUTE FORCE

LOGIN ATTEMPTS

DID

The attacker tried many username and password combinations to break into the Raspberry Pi using Telnet

FINDINGS

- Username: "root" → Password: "123456"
- Username: "enable" → Password: "system"
- Result: "Bad username/Password" repeatedly
- This is called a dictionary attack

WHY IS IT CRITICAL

This is why the attack happened. The device was using default passwords (like root:123456). If the owner had changed the password, the attacker could never get in. This was 100% preventable.

EVIDENCE 3

The image shows a Wireshark packet capture window titled "Wireshark · Follow TCP Stream (tcp.stream eq 5119) · 2018-05-09-192.168.100.103.pcap". The main pane displays the "User Access Verification" stream. The text in the stream is as follows:

```
.....  
User Access Verification  
Username:  
.....P.....root  
root  
Password:  
123456  
Bad username/Password  
Username:  
enable  
enable  
Password:  
system  
Bad username/Password  
Username:
```

Red circles highlight the following fields:

- Username:P.....root
- Password: 123456
- Username: enable
- Password: system

The bottom status bar indicates "8 client pkts, 7 server pkts, 12 turns." and "Entire conversation (277 bytes)". The "Show as" dropdown is set to "ASCII". The "Find" field is empty, and the "Find Next" button is highlighted.

On the right side, a packet list shows the following details:

- Seq=0 Ack=1 Win=8192 Len=0 MSS=1460 WS=1 T...
- Ack=1 Win=29248 Len=0 TSval=183076972 TSe...
- Do Terminal Type, Do Negotiate About Windo...
- Ack=55 Win=29248 Len=0 TSval=183076984 TS...
- d
- 5 Ack=4 Win=8192 Len=0 TSval=131560 TSecr=...
- ill Negotiate About Window Size, Suboption...
- 4 Ack=34 Win=8192 Len=0 TSval=131561 TSecr=...

The packet details pane shows the following fields:

- Destination
- Source
- Type
- Protocol Version 4
- Header Len...
- Differentiated Services Field

The bottom status bar indicates "Packets: 1686291 · Displayed: 24 (0.0%)"

IOC TABLE

IOC TYPE	VALUE	SIGNIFICANCE
Infected Device IP	192.168.100.103	Victim smart IoT device (Raspberry Pi) generating malicious outbound traffic.
Infected Device MAC	b8:27:eb:c7:e9:c9	Physical hardware address associated with the compromised Raspberry Pi.
Primary C2 Server IP	219.243.27.65	Active external Command & Control node interacting with the victim via Telnet.
Secondary Threat IP	71.181.6.40	Additional external host observed in concurrent unencrypted communication paths.
Target Protocol / Port	TCP Port 23 (Telnet)	TCP Port 23 (Telnet) Unsecured vector exploited for password dictionary attacks.

Event Timeline

TIME	EVENT
T+0 min	Attacker starts Telnet brute-force on Port 23
T+2 min	Attacker successfully logs in using "root:123456"
T+3 min	Device starts sending outbound traffic to C2 servers
T+10 min	Security team applies ACL to block attackers
T+11 min	All attacker traffic is blocked
T+12 min	Port security and VLAN isolation enabled

Mitigation: VLAN Segmentation

Problem: All devices were on the same flat network , attacker could reach everything

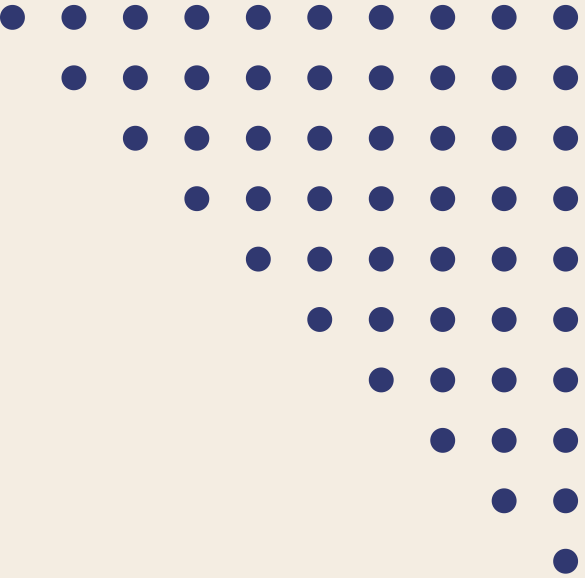
VLAN	Name	Ports	Purpose
VLAN 1	Management	Fa0/1–Fa0/5	Admin access only
VLAN 10	Internal Users	Fa0/6–Fa0/15	Normal user devices
VLAN 20	Quarantine	Fa0/16–Fa0/24	Isolated compromised device

RISK ASSESSMENT

Risk	Likelihood	Impact	Status	Mitigation Applied
Telnet Plaintext Interception	High	High	Resolved	SSH Replacement
Brute Force Login Attack	High	Critical	Resolved	Strong Passwords + Lockout Policy
C2 Server Communication	High	Critical	Resolved	ACL Blocking C2 IP
VLAN Hopping Attack	Medium	High	Resolved	Port Security + VLAN Pruning
Lateral Movement in Network	Medium	High	Resolved	Network Segmentation
Data Exfiltration	Low	Critical	Resolved	Traffic Monitoring + Firewall

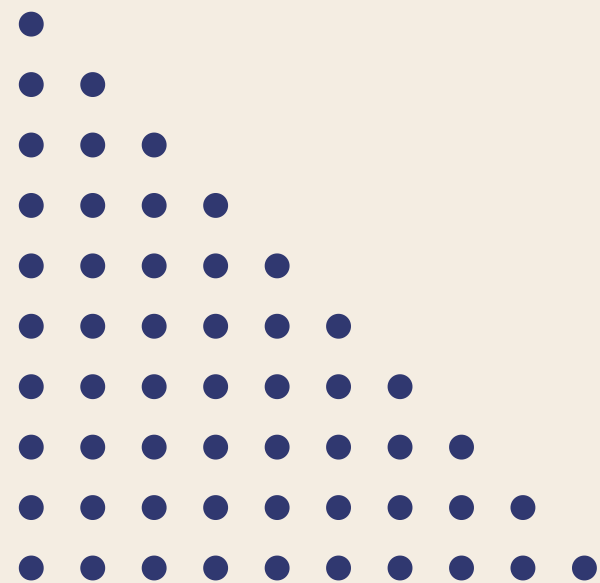
Final Security Test Results

Security Test Case ID	Operational Feature Tested	Expected Mitigation Outcome	Actual Security State	Test Status
TC-SEC-01	External C2 Port 23 Access	Firewall must explicitly block access	Connection Refused / Dropped	PASS
TC-SEC-02	Local Administrative Access	Enforced encryption via SSH Port 22	Encrypted Session Established	PASS
TC-SEC-03	Network Segregation Boundary	VLAN 20 must isolate IoT nodes	Local broadcast traffic locked inside	PASS
TC-SEC-04	Default Alphanumeric Dictionary	Reject standard common parameters	Device completely denies access	PASS



GITHUB REPOSITORY LINK

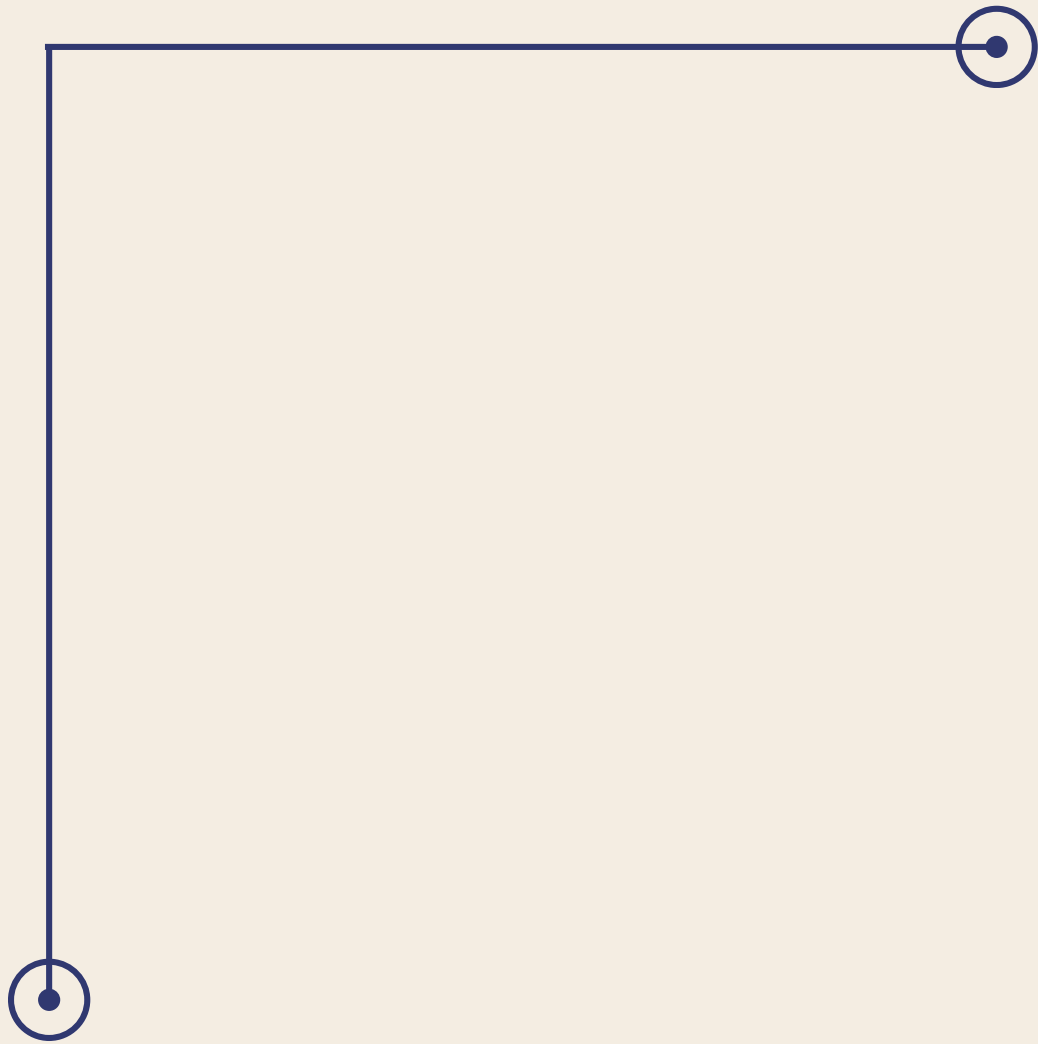
- <https://github.com/aribawaseemsiddiqui08/PCAP-Forensics-Problem7.git>



CONCLUSION

In conclusion, we found a Raspberry Pi device that was hacked because it was using default credentials. The attacker used Telnet to break in and then controlled the device remotely from a C2 server. We fixed everything by disabling Telnet, enabling SSH, blocking the attacker's IP, and isolating the device in VLAN 20. All our security tests passed. The network is now safe. The most important lesson here is simple: always change default passwords.





THANK YOU

